

ANALYSIS OF THE MAIN PHYSIOLOGICAL FACTORS IN CLIMBING PERFORMANCE



When it comes to training and performance, comparing bouldering to sport climbing is like comparing ping-pong to tennis. There are rackets, a ball and a net in both, but they're completely different sports. And it's the same in climbing: the demands of each type of climbing, and therefore the physiological profiles of each type of climber, are quite different. That said, with a well-designed training programme, the benefits of training specifically for one discipline can be transferred to other disciplines.

THE OLD PARADIGM OF PERFORMANCE: INTENSITY AND ENERGY SYSTEMS ACCORDING TO NUMBER OF MOVES

Sports physiology explores how the body responds to different stimulation in sport, laying the foundations for the development of sports training. For decades, there have been countless studies on all types of responses (metabolic, cardiovascular, and so on) in 'established' or 'popular' sports such as cycling, swimming and rowing. However, quality research into climbing has only really begun to emerge in recent years.

Until very recently, climbing has been likened to athletics, and this is still the case outside of competition and in popular culture. As such, the physiological demands and training methods of one have been applied to the other. A four-move boulder problem that takes 13 seconds to climb is compared to the 100 metres, so that's the anaerobic alactic system. A 20-move route that takes about a minute to climb is compared to the 400 metres, so that's anaerobic lactic. A 100-move route that takes 30 minutes to climb is compared to the 10K, so aerobic ... and so on. The dominance of one metabolic pathway over the others is based on the duration of activity and the number of moves, just like in athletics. And whatever training methods are used in athletics (for example, levels of intensity or intervals) are used in climbing as well. Epic fail.

Cyclic sports (like cycling) are based on repetitive movements that involve the concentric contraction of large muscle groups to produce a unified systematic response. Continuing with the example of cycling: the heart pumps lots of blood to the leg muscles, and the lungs help the metabolism to expel large amounts of CO₂ (a by-product of the aerobic energy system). When the glycogen stored in our muscles runs out, the liver provides its glycogen stores so that the cyclist can keep pedalling. The whole body works in unison to supply energy to the legs. If the legs can't keep producing energy, the cyclist will stop pedalling, even if their pecs or lats aren't fatigued.

In climbing, the main limiting factor is our forearm muscles. Failure to produce (and keep producing) enough force to keep our fingers in contact with a hold is the main reason we fall off (Vigouroux and Quaine, 2006). Our forearm muscles are much smaller than the legs of our cyclist. A small link in the chain, a bottleneck, is what limits performance. Our body's response to the energy demand of a small group of muscles is incomparable to the way in which it responds to the needs of the large muscle groups.

What's more, climbing involves intermittent isometric contractions at different intensities. In fact, it's practically impossible to calculate the intensity or force with which we grip a hold (even if it's fitted with a dynamometer), because even on the same hold, the amount of force will vary depending on body position (Michailov, 2014). This points to a model with very different needs and characteristics to cyclic sports.

A study on changes in lactate levels during the 400-metre sprint found lactate levels of around 17 mmol/L (Hirvonen *et al.*, 1992), and anaerobic strength (Wingate) tests on BMX bikers found lactate concentrations of 29 mmol/L (Zabala *et al.*, 2011). However, in sport climbers and boulderers, on a national and international level, in lab testing and official competitions, studies have found much lower concentrations of 5 to 7 mmol/L (Gáspari *et al.*, 2015; La Torre *et al.*, 2009; Billat *et al.*, 1995; Schöffl *et al.*, 2006). As lactate is simply a waste product of anaerobic glycolysis (so higher levels of lactate indicate greater predominance of this metabolic pathway in energy production), these findings show that anaerobic lactic metabolism is far from the predominant energy system in climbing.

Therefore, it makes little sense to approach forearm training, and training for climbing in general, as if we were runners or cyclists.

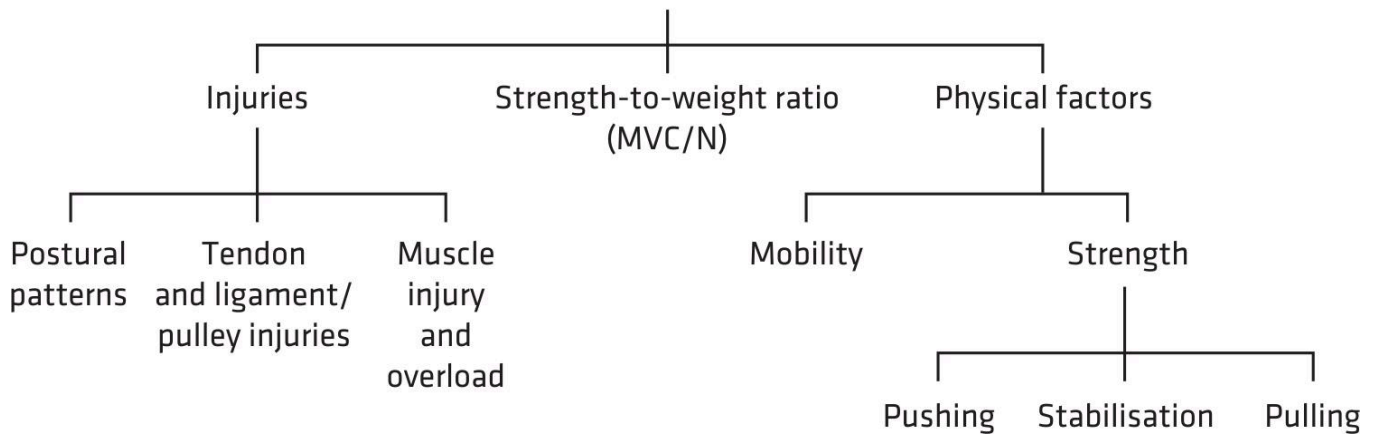
NEW FINDINGS ABOUT LIMITING FACTORS IN CLIMBING PERFORMANCE: ROUTES AND BOULDERING

So, what are the limiting factors in climbing performance? To answer this question, we should first distinguish between two different levels: general (factors that affect the whole body), and local (the crucial link: the forearm).

Let's begin at the general level. Broadly speaking, novices or intermediate climbers will be most affected by these factors (in elite climbers, these factors are optimised) and without a well-designed training plan they will have little to no control over these factors. As seen in the diagram overleaf, there are three main factors that influence our ability as climbers: physical factors, strength-to-weight ratio and injuries.

- **Physical factors:** factors that determine how we're able to move when climbing.
 - **Strength:** inevitably, strength is one of the factors that will either help or hinder our movement on the wall. While an elite climber can do pull-ups with lots of added weight or with just one arm, a novice might struggle to do even one regular pull-up. Likewise, the strength to stabilise our joints and core, in addition to overall balance, is crucial for certain moves like stopping a barn door. And, of course, the ability to push down and stand up (propulsion) on key footholds is crucial on any boulder problem or route.
 - **Mobility:** as discussed in [chapter 3](#), poor mobility will limit the positions we can get into on the wall, how far into certain positions we can get and how difficult it will be to move out of them.

GENERAL PERFORMANCE-LIMITING FACTORS



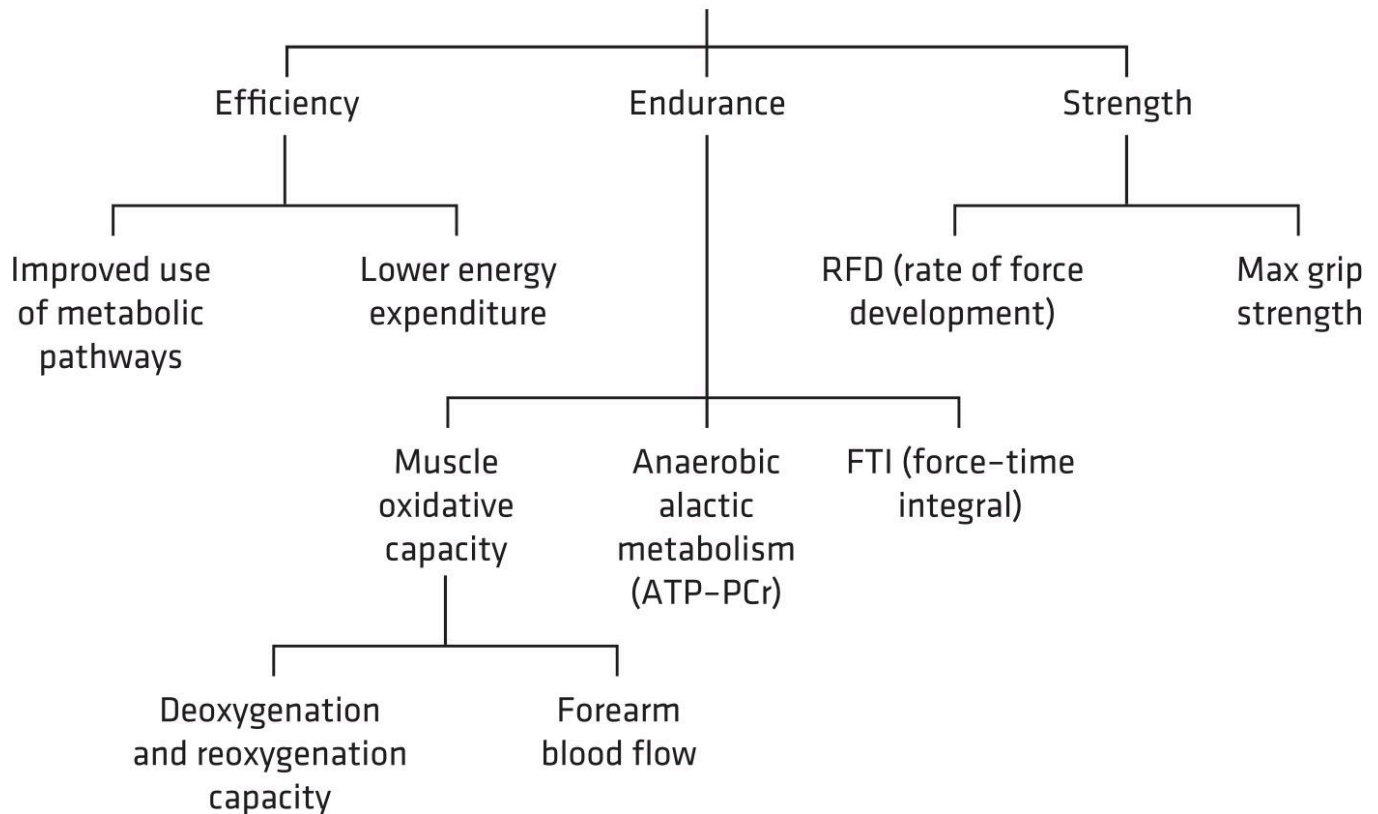
- **Strength-to-weight ratio:** as we can see from just looking at elite climbers and according to numerous studies including a PhD thesis by Couceiro (2010), on an anthropometric level, top-level climbers are not that heavy and, importantly, have a very low body fat percentage. Despite being the best energy store in the human body, fat is not contractile, so rather than helping us to climb it actually weighs us down, meaning it takes more effort and energy to get up the wall. It's not the total weight of a climber that's important, but rather their 'useful weight'. The more soft tissue we have that is useful/contractile/able to generate force, and the less additional weight, the better our strength-to-weight ratio. For example: take a 67-kilogram climber who can do a pull-up with 45 kilograms of added weight. In total, they're lifting 112 kilograms. Their strength-to-weight ratio is 1.67 kilograms per kilogram of body weight. That is, every kilogram of their body is able to generate enough force to lift 1.67 kilograms. Now take an 80-kilogram climber who can't do any unassisted pull-ups. They need the equivalent of 10 kilograms of help to do one pull-up. Their strength-to-weight ratio is 0.875 kilograms per kilogram of body weight. That is, every kilogram of their body is able to generate enough force to lift 0.875 kilograms, almost half that of the 67-kilogram climber. In scientific terms, this is known as (MVC/N): the force produced in a maximum voluntary contraction relative to the weight of the climber in newtons. Studies performed by Fryer *et al.* (2015b, 2015c) show that elite climbers have a much higher strength-to-weight ratio than advanced climbers, intermediate climbers and non-climbers.
- **Injuries:** regardless of how strong, lean or mobile we are, if we don't follow a well-designed and individualised training plan, we're putting ourselves at an exponential risk of getting injured and even having to give up climbing. Injury prevention covers a broad range of areas, such as the protection of soft tissue (mainly tendons and pulleys) through gradual progression and adaptation to training. It also covers the correction of postural adaptations or deformations caused by excessive tension in certain muscle groups. And finally, it includes agonist/antagonist muscle balance and the optimal (not excessive) loading of different muscle groups.

At this general level, we see the importance of the body as a whole (fingers of steel are of little use if you can't do a single pull-up or if you're nursing a shoulder injury). Now let's look at the latest research into the key limiting factor in climbing: the forearm.

- **Strength:** once again, strength takes centre stage. There are two factors that determine the importance of strength:
 - **Maximum grip strength:** Schöffl *et al.* (2006), Magiera *et al.* (2013), Ozimek *et al.* (2016) and López-Rivera (2009) suggest that maximum grip strength is one of the key deciding factors of climbing performance. And, as mentioned earlier, an inability to produce and/or keep producing the necessary force to remain on a handhold is the main reason for a failed ascent or fall (Vigouroux and Quaine, 2006). In turn, Fryer *et al.* (2017) found that boulderers have much higher MVC than sport climbers (so if you're a boulderer, you need to focus on finger strength training). Mermier *et al.* (2000) performed a comparative study on men and women and found that, considering strength relative to mass, men have much better grip strength than women. In addition, Ozimek *et al.* (2017) compared elite climbers (F8b/8c) with advanced climbers (F7c+/8a) and found that, for a three-second max-weight hang, the elite group could hang with 14 per cent more weight than the advanced group.
 - **RFD (rate of force development):** according to Fanchini *et al.* (2013), RFD is the deciding factor of performance in boulderers, as their RFD was found to be 36.73 per cent higher than in sport climbers. Laffaye *et al.* (2016) saw similar findings comparing boulderers to sport climbers. RFD is the amount of force developed per unit of time. In other words, it's how fast you can develop force. For climbing, the standard unit of time is 200 milliseconds. It's a bit like calculating acceleration: how much force you can produce in 200 milliseconds. This concept is closely linked to (but not the same as) explosive strength and the ability to do dynamic and explosive moves. It is also a crucial factor in contact strength. For example: imagine a climber doing a dynamic move to a sloper. To hold the sloper, they'll need to apply 200 newtons of force. Their MVC is 280 newtons so they should be able to stick the move. However, they have a very low RFD and can only generate 170 newtons when they hit the sloper, meaning they can't keep hold of it and they fall off. Levernier and Laffaye (2019) found that RFD_{200ms} is a variable that can

distinguish a climber's level of ability. As such, the ability to generate large amounts of force as quickly as possible, like making rapid and strong contact with a hold, is essential to reach a high level of performance.

LOCAL PERFORMANCE-LIMITING FACTORS (FOREARM)



- **Endurance:** the main physical quality that differentiates between sport climbers and boulderers (Fryer *et al.*, 2017) and the biggest limiting factor when it comes to sending a route (Michailov, 2014). The ability to maintain a muscle contraction or a series of contractions to exhaustion is what sets apart an elite climber from an advanced climber (Saul *et al.*, 2019). Interestingly, Philippe *et al.* (2012) compared time to exhaustion in male and female climbers for intermittent isometric contractions. Female climbers showed longer time to exhaustion, even at a lower climbing grade (both redpoint and on sight) than their male counterparts, although no concrete physiological explanation was found for this result. Endurance is determined by three things:
 - **Force-time integral (FTI):** a variable that shows the amount of force which can be maintained over a certain length of time (intermittent or continuous). It is calculated using the formula $0.4 \times \text{duration of the contraction (s)} \times \text{force (N)}$ and is measured in newtons per second (Fryer *et al.*, 2015b). More advanced climbers can do harder moves (moves that require more force) for longer. For example: picture a route with 25 moves. Each move requires 200 newtons of force. Let's assume that all climbers who attempt this route take three seconds to do each move. A novice climber can do five moves so their FTI will be 1,200 N·s. An intermediate climber can do 12 moves so their FTI will be 2,880 N·s. And an elite climber who can climb the whole route will have an FTI of 6,000 N·s. This variable is therefore directly proportional to elite, intermediate and novice levels and gives significantly different results for each group (Fryer *et al.*, 2015b). In order to measure FTI, you'd need to attach a dynamometer to a hold and perform a test to exhaustion (intermittent or continuous). The device will then provide a reading.



- *Anaerobic alactic metabolism*: as surprising as it may seem, the main pathway responsible for providing ATP to the forearms is the anaerobic alactic pathway and not anaerobic glycolysis (Bertuzzi *et al.*, 2007). Isometric contractions (duration of contact with each hold) during the crux section of a climb usually last for no more than 10 to 12 seconds (López-Rivera, 2014d), meaning that we are mainly using anaerobic alactic metabolism during this time. As the contractions are intermittent, with a (short) period of recovery between each hold, the aerobic system will try to replenish as much creatine phosphate as possible during this time to provide enough energy for the next move. We'll look at this in greater depth later on.
- *Muscle oxidative capacity*: the true crux of the matter. This refers to the ability to replenish energy. After a systematic review of over 350 studies, Saul *et al.* (2019) concluded that the oxidative capacity of the flexor digitorum profundus (FDP) is directly proportional to a climber's maximum redpoint grade. Oxidative capacity is dependent on:
 - *Forearm blood flow*: continuous high-intensity contractions, coupled with relatively negligible recovery between these contractions, can progressively limit blood flow to the forearm muscles. This, in turn, limits oxygen supply, the replenishment of energy reserves (Michailov, 2014; Staszkiwicz *et al.*, 2018) and the ability to flush out metabolic waste (Vigouroux and Quaine, 2006). However, in adaptation to training, more advanced climbers have developed wider arteries that increase blood flow to the forearm muscles (Fryer *et al.*, 2015a). They've also demonstrated a degree of vasodilation (widening of the blood vessels) during recovery, and their muscles, when contracting, exert less pressure on the arteries (metaboreflex), ensuring as much blood as possible gets to the muscles (Macleod *et al.*, 2007; Fryer *et al.*, 2015b; Thompson *et al.*, 2015; Green and Stannard, 2010). Another adaptation found to increase blood flow in elite climbers is known as capillary angiogenesis. This is the creation of new capillaries (blood vessels) that supply even more blood to the different muscle fibres (Macleod *et al.*, 2007; Philippe *et al.*, 2012; Fryer *et al.*, 2015a). Increased blood flow during recovery is directly linked to a higher FTI (Fryer *et al.*, 2015b), and elite climbers reoxygenate muscle 83 per cent faster than intermediate climbers (Fryer *et al.*, 2015a).
 - *Deoxygenation and reoxygenation capacity*: the biggest determining factor of a muscle's oxidative capacity and, as such, of forearm endurance (Fryer *et al.*, 2016). Elite climbers are much better at getting the oxygen carried in the bloodstream to their muscles (reoxygenation) and at removing oxygen from their muscles (deoxygenation), in both intermittent and continuous contractions. This means they can both use more and replenish more PCr stores to produce more energy (Fryer *et al.*, 2015a; Fryer *et al.*, 2015b; Fryer *et al.*, 2015c; Fryer *et al.*, 2017; Michailov, 2014; Green and Stannard, 2010; Philippe, *et al.*, 2012). In addition, greater deoxygenation during contractions and greater reoxygenation during recovery results in a higher FTI (Fryer *et al.*, 2015b). The deoxygenation and reoxygenation capacity of our muscles can be measured using NIRS (near infrared spectroscopy). There are several types of wearable NIRS devices but they're currently only designed for cyclic sports and not for climbing. Maybe in the future, we'll be able to use one a bit like a runner uses a heart rate monitor.
- *Efficiency*: efficiency can be defined as using less energy to perform the same action or task. Zigzagging between two points is effective, insofar as you leave one point and arrive at the other. However, going in a straight line is not only effective but also more efficient: using less energy to perform the same task. Bertuzzi

et al. (2007) suggest that for climbers, economy of movement is more important than a good metabolism. Efficiency is linked to technique, visualisation, good footwork, motor abilities and skills, psychological factors, and so on (Fryer *et al.*, 2016). Only continued and extensive practice, experience, and mileage on both plastic and rock will lead to better economy of movement. Efficiency in climbing can be measured through:

- *Lower energy expenditure:* Baláš *et al.* (2014b) compared energy expenditure in novice climbers and climbers who had climbed up to F8b. Both groups climbed a circuit of 15 moves, six times in a row; first at 90 degrees (vertical) and another five times immediately afterwards at 105 degrees (overhanging). The circuit was graded at around IV using the UIAA grading system (so, French grade 4a). The elite climbers used up to a fifth less energy than the novice climbers, as a result of better economy of movement. Another study by Baláš *et al.* (2014a) compared the FTI produced by the loading of footholds in novice and intermediate climbers, linking this data to the energy expenditure in each group. Climbers who put more weight through their feet (producing a higher FTI) had up to 11 per cent lower energy expenditure and 14 per cent lower heart rate.
- *Improved use of metabolic pathways:* acidosis, which causes a build-up of lactate in the forearm muscles, has been proven to cause fatigue (Philippe *et al.*, 2012). In a systematic review of over 350 studies, Saul *et al.* (2019) confirmed that elite climbers have lower concentrations of lactate than recreational climbers. This suggests that elite climbers are able to produce more energy via anaerobic alactic and aerobic pathways, enough for the hardest sections of a climb, without having to switch to the anaerobic lactic pathway, unlike climbers with a lower level of ability.